Lecture 21 DPP - Solutions

```
class Car:

    def __init__(self, brand):
        self.brand = brand

car1 = Car("BMW")
car2 = car1

car2.brand = "Audi"

print(car1.brand)
```

```
Audi
```

```
print(car2.brand)
```

```
Audi
```

```
class Movie:

    def __init__(self, name):
        self.name = name

movie = Movie("Avatar")

print(hasattr(movie, "rating"))
```

```
False
```

```
class Student:

    def __init__(self):
        self.age = 20

student = Student()

print(student.age + 5)
```

```
25
```

```
class Animal:

    def sound(self):
        print("Some Sound")

dog = Animal()

dog.sound()
dog.sound()
dog.sound()
```

```
Some Sound
Some Sound
Some Sound
```

```
class Laptop:

    def __init__(self, brand, ram):
        self.brand = brand
        self.ram = ram

laptop = Laptop("Dell", 16)

print(laptop.brand, laptop.ram)
```

```
Dell 16
```

```
class Book:

    def __init__(self, title):
        self.title = title

book1 = Book("Python")
book2 = Book("SQL")

print(book1.title)
print(book2.title)
print(book1.title == book2.title)
```

```
Python
SQL
False
```

```
class Student:

    def greet(self):
        return "Welcome"

student = Student()

message = student.greet()

print(message)
```

```
Welcome
```

```
class Bank:

    def __init__(self, balance):
        self.balance = balance

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount

account = Bank(1000)

account.withdraw(200)
account.withdraw(300)

print(account.balance)
```

```
500
```

Error Handling

```
print("Hello"
```

```
  File "/tmp/ipykernel_2098/1647630127.py", line 1
    print("Hello"
                ^
SyntaxError: incomplete input
```

Next steps: Explain error

```
Print("Hello")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_2098/244804411.py in <cell line: 0>()
----> 1 Print("Hello")

NameError: name 'Print' is not defined
```

Next steps: Explain error

```
if 10>5
    print("Hello")
```

```
  File "/tmp/ipykernel_2098/4088754853.py", line 1
    if 10>5
          ^
SyntaxError: expected ':'
```

Next steps: Explain error

```
if 10>5:
print("Hello")
```

```
  File "/tmp/ipykernel_2098/350778045.py", line 2
    print("Hello")
    ^
IndentationError: expected an indented block after 'if' statement on line 1
```

Next steps: Explain error

```
a = 10
b = 0

print(a/b)
```

```
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
/tmp/ipykernel_2098/1256220760.py in <cell line: 0>()
      2 b = 0
      3
----> 4 print(a/b)

ZeroDivisionError: division by zero
```

Next steps: Explain error

```
a = 0
b = 10

print(a/b)
```

```
0.0
```

```
a = [3,7,5]
print(a[4])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
/tmp/ipykernel_2098/987877489.py in <cell line: 0>()
      1 a = [3,7,5]
----> 2 print(a[4])

IndexError: list index out of range
```

Next steps: Explain error

```
age = int(input("Enter the age:"))
```

```
Enter the age:aditya
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_2098/1365288385.py in <cell line: 0>()
----> 1 age = int(input("Enter the age:"))

ValueError: invalid literal for int() with base 10: 'aditya'
```

Next steps: Explain error

```
#Write a code to take two numbers as input from the user and print their division.

a = int(input())
b = int(input())

print(a/b)
```

```
4
6
0.6666666666666666
```

```
a = int(input())
b = int(input())

print(a/b)
```

```
6
3
2.0
```

```
a = int(input())
b = int(input())

print(a/b)
```

```
5
0
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
/tmp/ipykernel_2098/2956846593.py in <cell line: 0>()
      2 b = int(input())
      3
----> 4 print(a/b)

ZeroDivisionError: division by zero
```

Next steps: Explain error

```
try:
  n = int(input())
  print(100/n)
except:
  print("Some error has occured")
```

```
5
20.0
```

```
try:
  n = int(input())
  print(100/n)
except:
  print("Some error has occured")

print("Hello")
```

```
0
Some error has occured
Hello
```

```
n = int(input())
print(100/n)
print("Hello")
```

```
0
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
/tmp/ipykernel_2098/1489522472.py in <cell line: 0>()
      1 n = int(input())
----> 2 print(100/n)
      3 print("Hello")

ZeroDivisionError: division by zero
```

Next steps: Explain error

```
try:
  n = int(input())
  print(100/n)
except:
  print("Cannot divide by 0")
```

```
0
Cannot divide by 0
```

```
try:
  n = int(input())
  print(100/n)
except:
  print("Cannot divide by 0")
```

```
aditya
Cannot divide by 0
```

```
try:
  a = [4,5,7]
  print(a[6])
except:
  print("Invalid index")
```

```
Invalid index
```

Catching Specific Errors

```
try:
  n = int(input())
  print(100/n)

except Exception as e:
  print(e)
```

```
aditya
invalid literal for int() with base 10: 'aditya'
```

```
try:
  n = int(input())
  print(100/n)

except Exception as e:
  print(e)
```

```
0
division by zero
```

```
try:
  n = int(input())
  print(100/n)

except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
0
Division by 0 is not allowed
```

```
try:
  n = int(input())
  print(100/n)

except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
aditya
Please enter numbers only
```

```
try:
  n = int(input())
  print(100/n)

except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
try:
  n = int(input())
  print(100/n)

except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
try:
  n = int(input())
  print(100/n)
  print("Hello"
except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
except Exception as e:
  print("Some error has occured!")
```

```
  File "/tmp/ipykernel_2098/1470257123.py", line 4
    print("Hello"
          ^
SyntaxError: '(' was never closed
```

Next steps: Explain error

```
try:
  n = int(input())
  print(100/n)
  a = [3,6]
  print(a[4])
except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
except Exception as e:
  print("Some error has occured!")
```

```
5
20.0
Some error has occured!
```

```
try:
  n = int(input())
  print(100/n)
  a = [3,6]
  print(a[4])
except Exception as e:
  print("Some error has occured!")
except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
0
Some error has occured!
```

```
try:
  n = int(input())
```

```
  print(100/n)
  a = [3,6]
  print(a[4])
except Exception as e:
  print("Some error has occured!")
except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
adit
Some error has occured!
```

```
try:
  n = int(input())
  print(100/n)
  a = [3,6]
  print(a[4])
except Exception as e:
  print("Some error has occured!")
except ZeroDivisionError:
  print("Division by 0 is not allowed")
except ValueError:
  print("Please enter numbers only")
```

```
5
20.0
Some error has occured!
```

```
#else block runs only when no exception occurs

try:
    num = int(input())

except ValueError:
    print("Invalid input")

else:
    print("You entered:", num)
```

```
5
You entered: 5
```

```
#else block runs only when no exception occurs

try:
    num = int(input())

except ValueError:
    print("Invalid input")

else:
    print("You entered:", num)
```

```
adit
Invalid input
```

```
#finally block always executes irrespective whether there is an error or not

try:
    num = int(input())

except ValueError:
    print("Invalid input")

finally:
    print("Program finished")
```

```
4
Program finished
```

```
#finally block always executes irrespective whether there is an error or not

try:
```

```
    num = int(input())

except ValueError:
    print("Invalid input")

finally:
    print("Program finished")
```

```
adit
Invalid input
Program finished
```

Quiz

```
try:
    print(10 / 2)
except:
    print("Error")
```

```
5.0
```

```
try:
    print(10 / 0)
except:
    print("Cannot divide")
```

```
Cannot divide
```

```
numbers = [10, 20, 30]

print(numbers[5])
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
/tmp/ipykernel_2098/1432885474.py in <cell line: 0>()
      1 numbers = [10, 20, 30]
      2
----> 3 print(numbers[5])

IndexError: list index out of range
```

Next steps: Explain error

```
student = {
    "name": "Rahul"
}

print(student["age"])
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_2098/4291066776.py in <cell line: 0>()
      3 }
      4
----> 5 print(student["age"])

KeyError: 'age'
```

Next steps: Explain error

```
try:
    print("Python")
except:
    print("Error")
finally:
    print("Done")
```

```
Python
Done
```

```
try:
    print("Python")
except:
    print("Error")
else:
    print("Done")
```

```
Python
Done
```

```
try:
    n= int("a")
except:
    print("Error")
else:
    print("Done")
```

```
Error
```

```
# What is the purpose of the finally block?

# A. Runs only when there is no exception

# B. Runs only when an exception occurs

# C. Always executes

# D. Stops the program


# ans:C
```

```
try:
    print(100 / 25)
except:
    print("Error")
else:
    print("Success")
```

```
4.0
Success
```

```
try:
    x = 5
    print(x)
except Exception:
    print("Error")
```

```
5
```

```
try:
    a = 20
    b = 0
    result = a / b
    print(result)

except ZeroDivisionError:
    print("Cannot divide")

finally:
    print("Program Finished")
```

```
Cannot divide
Program Finished
```

DPP - Questions

```
try:
    numbers = [5, 10, 15]

    index = 2
```

```
    print(numbers[index])

except IndexError:
    print("Wrong Index")

else:
    print("Successful")

finally:
    print("Done")
```

```
try:
    age = int("Twenty")

    print(age)

except ValueError:
    print("Invalid Age")

finally:
    print("End")
```

```
try:
    student = {
        "name": "Rahul",
        "marks": 90
    }

    print(student["age"])

except KeyError:
    print("Key Missing")

print("Program End")
```

```
try:
    marks = 120

    if marks > 100:
        raise ValueError("Marks Invalid")

    print(marks)

except ValueError as e:
    print(e)

print("Completed")
```

```
try:
    num = int("25")

    print(100 / num)

except ValueError:
    print("Value Error")

except ZeroDivisionError:
    print("Division Error")

else:
    print("Calculation Done")
```

```
try:
    numbers = [2, 4, 6]

    print(numbers[5])

except IndexError:
    print("Index Error")

except Exception:
    print("General Error")

finally:
```

```
    print("Finished")
```

```
try:
    value = int("50")

    print(value)

except Exception:
    print("Error")

else:
    print("Success")

finally:
    print("Done")
```